

Edge AI Computing for Grape Leaf Disease Detection

Thesis Presentation



UNIVERSITÀ DI PISA

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Advisories:

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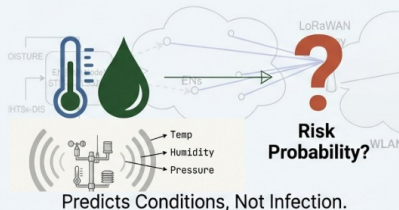
Dr. Palla Fabrizio

Prof. Zinnai Angela

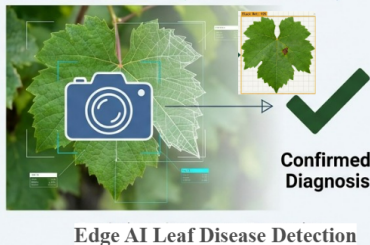
Candidate: *Gebrelibanos Tsegay*

Motivation → Beyond Environmental Data

Environmental Data Monitoring

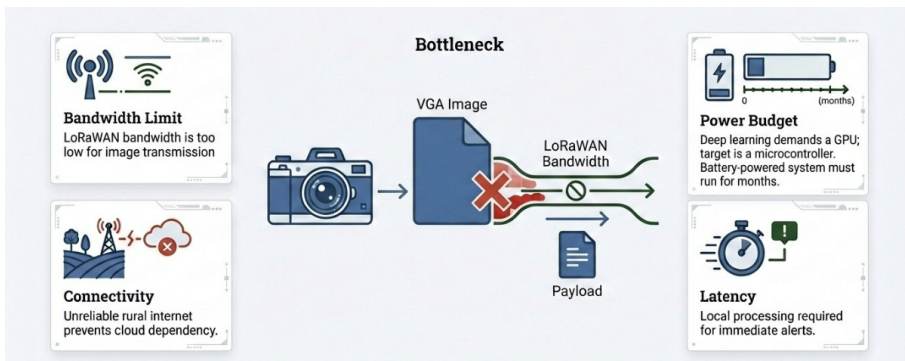


Thesis Goal (Direct Visual AI)



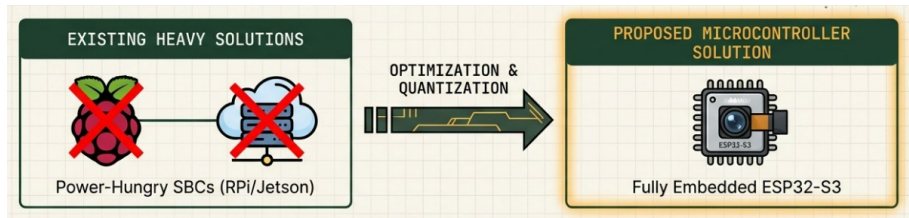
- *LoRaWAN-based WSNs are widely used for vineyard monitoring*
- *Environmental data predicts disease risk but doesn't confirm presence*
- *Visual-based edge detection confirms disease presence*

Problem Statement and Constraints



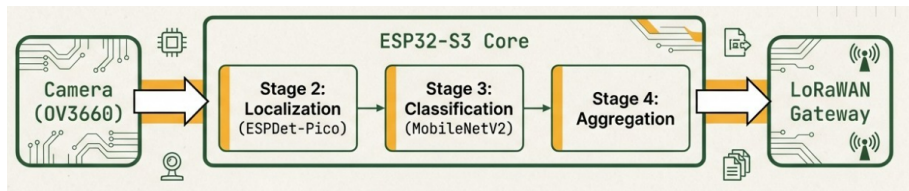
- *LoRaWAN Bandwidth and Power Constraints*
- *Image transmission over LoRaWAN is not feasible*
- *Battery-powered Sensors & Microcontrollers*

Research Gap and Proposed Solution



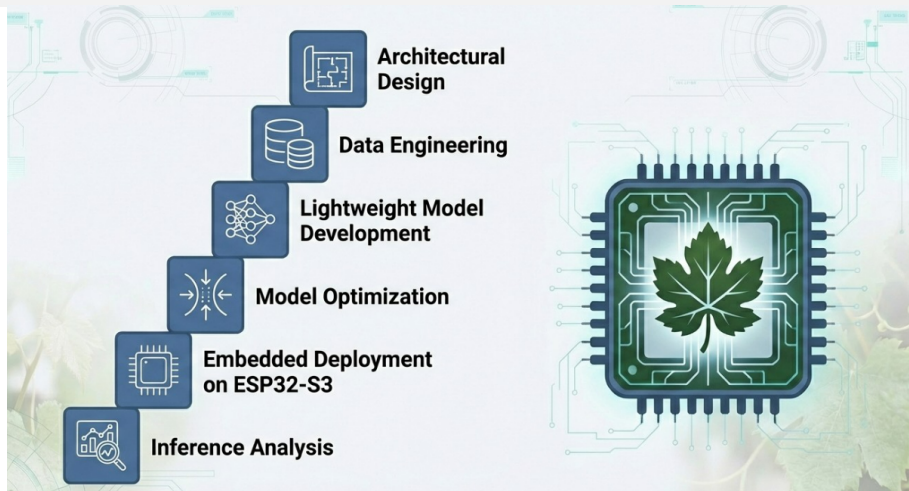
- *Lightweight Edge AI for Leaf Disease Detection*
- *On-device Inference*
- *Low-power AI-supported Microcontroller (ESP32-S3)*

Overall System Architecture



- **Stage1:** *Image Acquisition (VGA 640x480)*
- **Stage2:** *Leaf Localization (Optimized-YOLOv11)*
- **Stage3:** *Disease Classification (MobileNetV2) on cropped leaves*
- **Stage4:** *Result aggregation and LoRaWAN Transmission*

Methodology



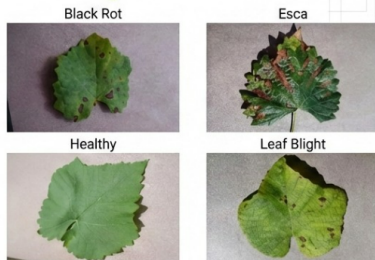
Dataset Preparation and Model Selection

Stage 1: Localization



Localization Dataset: complex vineyard images (Annotated 'Leaf').

Stage 2: Classification

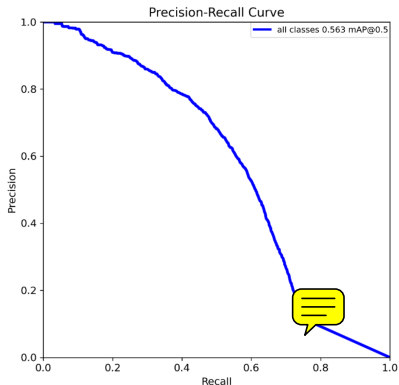


Classification Dataset: PlantVillage (images, 4 classes).

- **Why Split?** *Separation allows independent optimization on detection and classification tasks*

Leaf Localization

- **Model:** ESPDet-Pico (Custom YOLO Optimization)
- **Objective:** Locate and crop leaves in complex backgrounds
- **Performance:** 56.4% mAP@50 (Sufficient for ROI Extraction)



Disease Classification Model

- *MobileNetV2 architecture*
- *Input: 128×128 pixels*
- *Fine-tuning training strategy*
- *4 disease classes*

Overall Performance

Test Accuracy: 99.75%

Macro F1-score: 0.9980

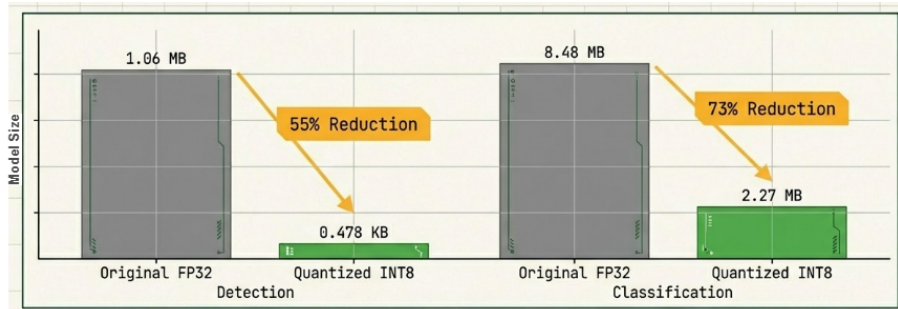
Precision: 0.9982

Recall: 0.9979

Class	Prec.	Recall	F1
Black Rot	1.0000	0.9915	0.9957
Esca	0.9928	1.0000	0.9964
Healthy	1.0000	1.0000	1.0000
Leaf Blight	1.0000	1.0000	1.0000
Macro avg	0.9982	0.9979	0.9980

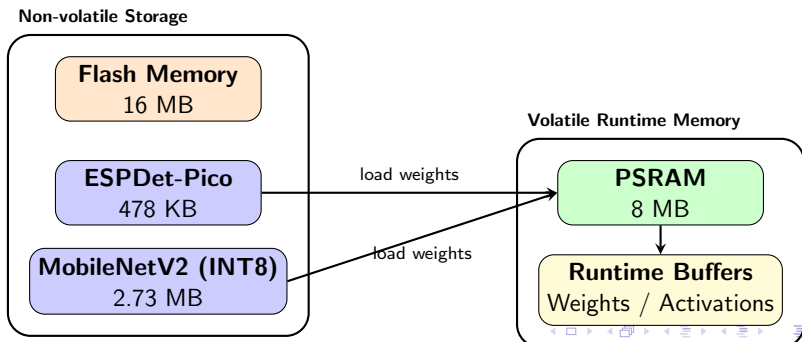
Model Optimization for Edge Deployment

- *Post-Training Optimization (PTQ) via ESP-PPQ (INT8)*
- *Leaf detection compression 1.06 M $\rightarrow$ 478 KB*
- *Disease classification compression 8.48 M $\rightarrow$ 2.27 MB*
- *Negligible accuracy loss ($<0.3\%$ drop)*
- *Convert into ESP-DL Compatible Model (.espd)*

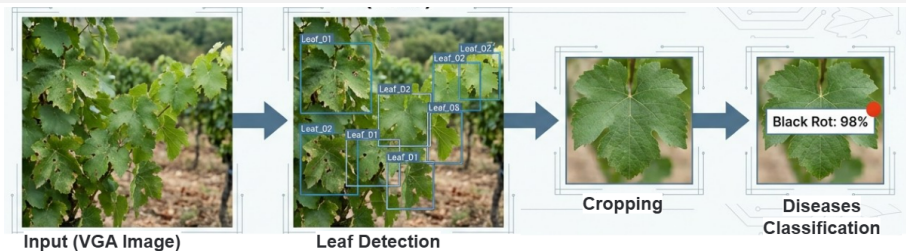


Edge AI Deployment on ESP32-S3

- **ESP32-S3** (*Xtensa LX7, 512 KB RAM, 8 MB PSRAM, 16 MB flash*)
- *Models embedded in flash*
- *Weights loaded into PSRAM at runtime*
- *All Runtime buffers allocated in PSRAM*

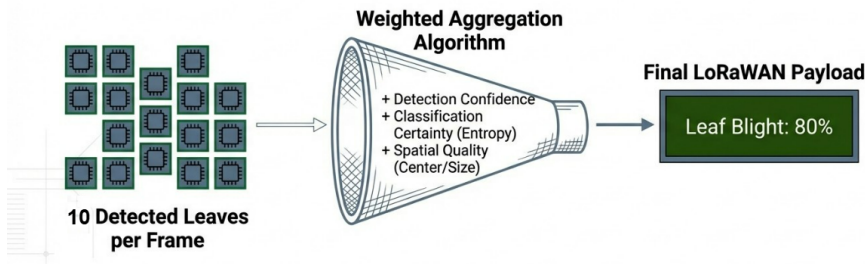


Inference Flow Design



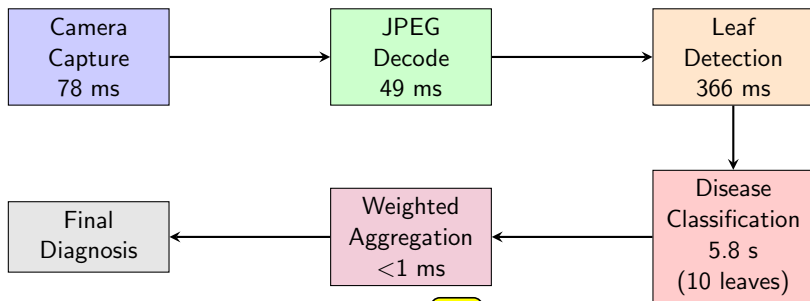
- **Two-Model Inference Pipeline**
- *Leaf Detection (Optimized-YOLOv11) on full VGA frames*
- *Disease Classification (MobileNetV2) on cropped leaf images*

LoRaWAN Payload and Aggregation Logic



- *Full frame prediction is aggregated from multiple leaf predictions*
- *The aggregation logic is weighted by leaf information (detection confidence, classification entropy, spatial quality)*
- *This aggregated inference result transmits over LoRaWAN*

Inference Performance



Total Pipeline Time: ~6.3 seconds per frame

- *This inference cycle performs every 6 hours (duty cycle)*
- *ESP32-S3 stays in deep sleep mode for log time (to save battery)*
- *With this duty cycle, 2400mAH battery can last for 10 months*

Conclusion and Future Work

- **Conclusion:** *Successfully demonstrated fully embedded visual leaf disease detection on MCU*
- **Key Contributions:** *Validate the inference pipeline(Capture → Detect → Crop → Classify) on ESP32-S3*
- **Future Work:** *Field testing, model optimization, and expanding to more diseases*

Thank You

Questions?